EIGHT EQUIVALENT DESCRIPTIONS OF PROJECTIVE MORPHISMS OVER A BASE WITH AN AMPLE INVERTIBLE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 14027-14045 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 14047-14100, the complete author proof (52 body lines).

Editorial note. Navigation stubs: 11 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Hub lemma of the chapter's projective/quasi-projective series; complements L16072 (quasi-projective) without overlap.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The quasi-projective characterization card (same chapter, collected as L16072), the Morphisms chapter dictionary between projective/H-projective/quasi-projective notions, and the relative Proj machinery (Divisors, lemmas relative-proj-projective and closed-subscheme-proj; Constructions, lemma-projective-space-bundle).

Source reference. Original lemma 01OE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14100–14100. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0801: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14100–14100. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B3U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14100–14100. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B42: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13840–13853. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01W8: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14100–14100. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VW: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14100–14100. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14100–14100. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01W9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14100–14100. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14100–14100. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BCL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14100–14100. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087S: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14100–14100. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme which has an ample invertible sheaf. Let $f : X \to S$ be a morphism of schemes. The following are equivalent*

(1) *$X \to S$ is projective,*
(2) *$X \to S$ is H-projective,*
(3) *$X \to S$ is quasi-projective and proper,*
(4) *$X \to S$ is H-quasi-projective and proper,*
(5) *$X \to S$ is proper and X has an ample invertible sheaf,*
(6) *$X \to S$ is proper and there exists an f-ample invertible sheaf,*
(7) *$X \to S$ is proper and there exists an f-very ample invertible sheaf,*
(8) *there is a quasi-coherent graded $\mathcal{O}_S$-algebra $\mathcal{A}$ generated by $\mathcal{A}_1$ over $\mathcal{A}_0$ with $\mathcal{A}_1$ a finite type $\mathcal{O}_S$-module such that $X = \underline{Proj}_S(\mathcal{A})$.*

Proof. Observe first that in each case the morphism f is proper, see Morphisms, Lemmas 01W9 and 01WC. Hence it suffices to prove the equivalence of the notions in case f is a proper morphism. We will use this without further mention in the following.

The equivalences (1) $\Leftrightarrow$ (3) and (2) $\Leftrightarrow$ (4) are Morphisms, Lemma 0BCL.

The implication (2) $\Rightarrow$ (1) is Morphisms, Lemma 01W9.

The implications (1) $\Rightarrow$ (2) and (3) $\Rightarrow$ (4) are Morphisms, Lemma 087S.

The implication (1) $\Rightarrow$ (7) is immediate from Morphisms, Definitions 01W8 and 01VM.

The conditions (3) and (6) are equivalent by Morphisms, Definition 01VW.

Thus (1) – (4), (6) are equivalent and imply (7). By Lemma 0B42 conditions (3), (5), and (7) are equivalent. Thus we see that (1) – (7) are equivalent.

By Divisors, Lemma 0B3U we see that (8) implies (1). Conversely, if (2) holds, then we can choose a closed immersion

$$i : X \longrightarrow \mathbf{P}^n_S = \underline{\mathrm{Proj}}_S(\mathcal{O}_S[T_0, \ldots, T_n]).$$

See Constructions, Lemma 01OE for the equality. By Divisors, Lemma 0801 we see that X is the relative Proj of a quasi-coherent graded quotient algebra $\mathcal{A}$ of $\mathcal{O}_S[T_0, \ldots, T_n]$. Then $\mathcal{A}$ satisfies the conditions of (8). □